```
USE intership_database;

CREATE TABLE Department (
dept_id INT PRIMARY KEY,dept_name VARCHAR(50));

CREATE TABLE Employee (
emp_id INT PRIMARY KEY AUTO_INCREMENT,emp_name VARCHAR(50),salary DECIMAL(10,2),hire_date DATE,dept_id INT,FOREIGN KEY (dept_id) REFERENCES Department(dept_id));

-- Department Data
INSERT INTO Department VALUES (10, 'IT'), (20, 'HR'), (30, 'Sales');

-- Employee Data
INSERT INTO Employee (emp_name, salary, hire_date, dept_id) VALUES
('Amit Sharma', 90000.00, '2026-01-15', 10),
('Priya Rai', 85000.00, '2025-11-20', 10),
('Rohan Das', 60000.00, '2026-04-10', 20),
('Neha Gupta', 95000.00, '2025-08-05', 10),
('Vikram Singh', 45000.00, '2026-03-01', 30),
('Anjali Verma', 60000.00, '2026-05-12', 20),
('Amit Sharma', 90000.00, '2026-01-15', 10);

# 1. Top 5 highest salary employees
SELECT * FROM Employee ORDER BY salary DESC LIMIT 5;
```

Result Grid | Filter Rows: | Edit:

emp_id	emp_name	salary	hire_date	dept_id
11	Neha Gupta	95000.00	2025-08-05	10
8	Amit Sharma	90000.00	2026-01-15	10
9	Priya Rai	85000.00	2025-11-20	10
10	Rohan Das	60000.00	2026-04-10	20
13	Anjali Verma	60000.00	2026-05-12	20
NULL	NULL	NULL	NULL	NULL

```
# 2. Department wise employee count
SELECT d.dept_name, COUNT(e.emp_id) AS total_employees
FROM Employee e
JOIN Department d ON e.dept_id = d.dept_id
GROUP BY d.dept_name;
```

Result Grid | Filter Rows:

dept_name	total_employees
IT	3
HR	2
Sales	1

 # 3. Find Second highest salary

```sql
SELECT MAX(salary) FROM Employee
WHERE salary < (SELECT MAX(salary) FROM Employee);
```

MAX(salary)
90000.00

 # 4. Employees whose salary > department average salary

```sql
SELECT e.emp_name, e.salary, e.dept_id
FROM Employee e
WHERE e.salary > (SELECT AVG(salary) FROM Employee WHERE dept_id = e.dept_id);
```

emp_name	salary	dept_id
Neha Gupta	95000.00	10

 # 5. Inner Join

```sql
SELECT e.emp_name, d.dept_name, e.salary
FROM Employee e
INNER JOIN Department d ON e.dept_id = d.dept_id;
```

emp_name	dept_name	salary
Amit Sharma	IT	90000.00
Priya Rai	IT	85000.00
Rohan Das	HR	60000.00
Neha Gupta	IT	95000.00
Vikram Singh	Sales	45000.00
Anjali Verma	HR	60000.00

#6. Left Join

```sql
SELECT e.emp_name, d.dept_name
FROM Employee e
LEFT JOIN Department d ON e.dept_id = d.dept_id;
```

emp_name	dept_name
Amit Sharma	IT
Priya Rai	IT
Rohan Das	HR
Neha Gupta	IT
Vikram Singh	Sales
Anjali Verma	HR

 # 7. Group By with Having

```
SELECT dept_id, AVG(salary) AS avg_salary
FROM Employee
GROUP BY dept_id
HAVING AVG(salary) > 65000;
```

Result Grid | Filter Rows:

dept_id	avg_salary
10	90000.000000

 # 8. Employees hired in last 6 months

```
SELECT emp_name, hire_date FROM Employee
WHERE hire_date >= DATE_SUB('2026-06-25', INTERVAL 6 MONTH);
```

Result Grid | Filter Rows:

emp_name	hire_date
Amit Sharma	2026-01-15
Rohan Das	2026-04-10
Vikram Singh	2026-03-01
Anjali Verma	2026-05-12

 # 9. Find the Duplicates records

```
SELECT emp_name, salary, dept_id, COUNT(*)
FROM Employee
GROUP BY emp_name, salary, dept_id
HAVING COUNT(*) > 1;
```

Result Grid | Filter Rows:

emp_name	salary	dept_id	COUNT(*)

 # 10. How to remove the duplicates record

```
DELETE e1 FROM Employee e1
INNER JOIN Employee e2
WHERE e1.emp_id > e2.emp_id
  AND e1.emp_name = e2.emp_name
  AND e1.salary = e2.salary;
```